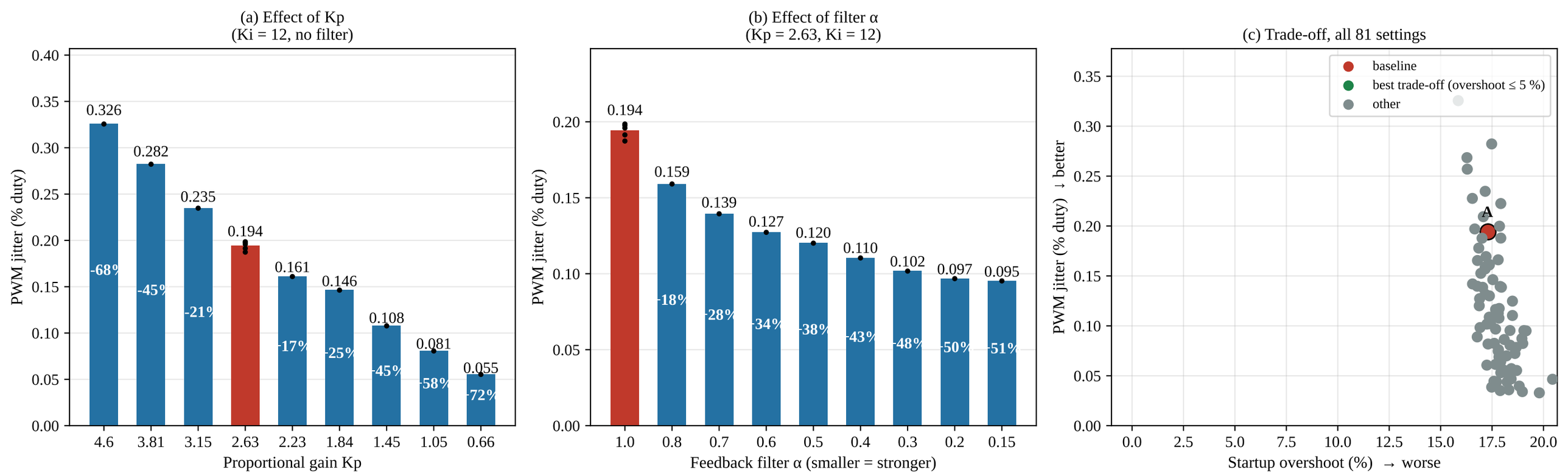
(a) Effect of Kp
(Ki = 12, no filter)
PWM jitter (% duty)
0.326
0.282
0.235
0.194
0.161
0.146
0.108
0.081
0.055
-68%
-45%
-21%
-17%
-25%
-45%
-58%
-72%
4.6
3.81
3.15
2.63
2.23
1.84
1.45
1.05
0.66
Proportional gain Kp
(b) Effect of filter α
(Kp = 2.63, Ki = 12)
PWM jitter (% duty)
0.194
0.159
0.139
0.127
0.120
0.110
0.102
0.097
0.095
-18%
-28%
-34%
-38%
-43%
-48%
-50%
-51%
1.0
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.15
Feedback filter α (smaller = stronger)
(c) Trade-off, all 81 settings
baseline
best trade-off (overshoot ≤ 5 %)
other
A
PWM jitter (% duty) ↓ better
Startup overshoot (%) → worse